

Reframe: Recording and Editing Character Motion in Virtual Reality

Qian Zhou
qian.zhou@autodesk.com
Autodesk Research
Canada

Aniruddha Prithul
aniruddha.prithul@autodesk.com
Autodesk Research
Canada

Hans Kellner
hans.kellner@autodesk.com
Autodesk Research
United States of America

Brian Pene
brian.pene@autodesk.com
Autodesk Research
United States of America

David Ledo
david.ledo@autodesk.com
Autodesk Research
Canada

Sebastian Herrera
sebastian.herrera@autodesk.com
Autodesk Research
United States of America

Hilmar Koch
hilmar.koch@autodesk.com
Autodesk Research
United States of America

George Fitzmaurice
george.fitzmaurice@autodesk.com
Autodesk Research
Canada

Fraser Anderson
fraser.anderson@autodesk.com
Autodesk Research
Canada



Figure 1: Reframe is an animation authoring interface that allows novice users to (1) choose a character, (2-3) record the performance, and (4) edit the captured motion in Virtual Reality.

ABSTRACT

Creating lifelike 3D character animations is traditionally complex and requires substantial skill and effort. To overcome this challenge, we introduce Reframe, a Virtual Reality animation authoring interface that allows users to record and edit motion. Reframe utilizes tracking technology in Virtual Reality headsets to capture the user's full-body motion, facial expressions, and hand gestures. To facilitate the editing process, we have developed an immersive motion editing interface that combines spatial and temporal control for character animation. This system extracts keyposes from the 3D character animation and displays them along a timeline, connecting the joints through 3D trajectories to depict the character's movement. We have created a proof-of-concept prototype that demonstrates how a single user can select and animate multiple characters in a scene. This system offers an interactive experience that explores the possibilities of future immersive animation technologies.

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1 INTRODUCTION

3D animation is becoming increasingly popular with individuals without professional expertise, and VTubers, streamers, and content creators are embracing this medium more frequently. However, the process of creating a 3D animation is complicated and requires substantial skill to achieve a realistic and convincing outcome. This complexity can be challenging for novice users, particularly in the realm of character animation with numerous joints and interconnected moving parts. To address this challenge, previous work have examined the use of Virtual Reality (VR) to enhance the process of authoring and editing animations. Immersive animation tools utilize the 6 Degrees-of-Freedom (DoF) tracking of controllers and hands to animate objects through puppeteering. Users can grab an object or joint using controllers or hands, move it to different positions at different time points, and record the resulting motion path [Lamberti et al. 2020]. The recent immersive approach of capturing full-body

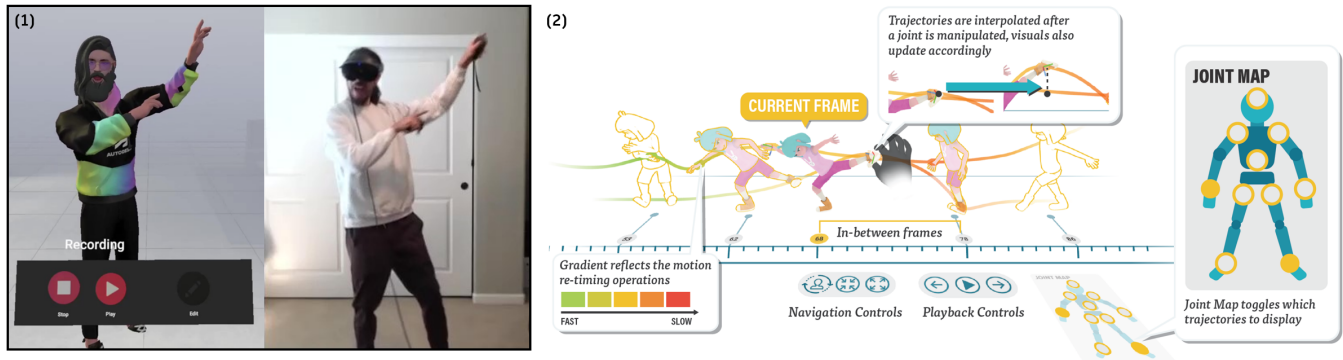


Figure 2: (1) The recording interface includes a mirrored character and supportive UIs. (2) The editing interface includes the main character, KeyPoses, Trajectories, and supportive UIs.

motion in VR [Ahuja et al. 2021] offers promise for using the user’s natural movement as a starting point. While capturing motion has become more accessible, editing the captured data remains challenging. Motion-capture systems continuously record a performer’s actions, producing a pose for each captured frame, rather than focusing on specific moments in time. This approach generates a large volume of data without high-level structures that describe the motion’s properties, making it time-consuming to edit the motion across frames. This highlights the need for a user-friendly editing experience with a self-contained VR interface for recording and editing, particularly catering to novice users.

We introduce Reframe, an approachable animation authoring interface designed for recording and editing motion in VR for novice users. Users can capture their full-body motion, facial expressions, and hand gestures. Furthermore, they can edit the captured motion in VR using a novice-friendly motion editing interface. Different from existing immersive animation interfaces, the editing interface employs Trajectories and KeyPoses to represent the motion, integrating spatial and temporal control and providing an approachable editing interface for novice users.

2 INTERFACE AND EXPERIENCE

Reframe system includes a motion recording interface and a motion editing interface in VR (Figure 2): **The recording interface** includes a mirrored character and supporting interactions. The mirrored character reflects a user’s movement one-to-one to enable visualizing their motion, facial expressions, and hand gestures. They can press the record button to start or stop recording (Figure 2.1). They can replay their recorded motion by pressing the play button. They can re-record the motion until they are satisfied with their clip. **The editing interface** [Zhou et al. 2024] is composed of a main character, KeyPoses, Trajectories, and supporting interfaces. KeyPoses are carefully calculated representative poses that offer a concise overview of the motion. Trajectories are 3D animation curves that traverse the joints of the KeyPoses, aiding users in visualizing the motion that occurs between them with the colour representing the re-timing operations. Since motion-captured data lacks high-level controls such as keyframes, KeyPoses are automatically extracted by analyzing changes in the Trajectories. Surrounded by KeyPoses, the main character rendered with solid color is positioned in the middle of the

tunnel that represents to the current frame (Figure 2.2). Users can select and drag it to scrub the time axis and navigate through the frames. Users can manipulate its pose by selecting and moving the joints. We use a full-body Inverse-Kinematics (IK) approach to support posing with 10 effectors (hands, feet, upper arms, upper legs, head, and spine). To help users select the joint, a spherical widget with local axes is displayed for each activated joint. Users can move and rotate the widget to control the pose in the current frame. The changes made in the current frame are propagated to adjacent frames bounded by a pair of the closest KeyPoses (in transparent color) on both sides of the main character. KeyPoses can also be used for re-timing a segment of motion. Users can select a left KeyPose as the start frame and a right KeyPose as the end frame to specify the range of motion that will be impacted by the modification. To modify the timing of the segment, users can adjust the length of Trajectories between the two KeyPoses by either shrinking or expanding them to speed up or down the motion. This re-timing process includes an ease-in-out function applied to the selected segment for a seamless speed adjustment.

As a demonstration of the proposed workflow, we developed a proof-of-concept prototype that runs on an untethered standalone VR headset. During the demo, attendees can first select a character that they would like to play (Figure 1.1). They can capture their performance, including full-body motion, facial expressions, and hand gestures while watching a mirrored character performing the same movement (Figure 2.1). They can edit the captured motion in VR using advanced 3D interaction techniques such as pointing, pinching, poking, grabbing, and various bimanual interactions (Figure 2.2). Together, the proposed approach enables new ways for novice users to create and edit character animations quickly in VR. We believe this demonstration provides a highly interactive and fun experience for attendees to author character animations live.

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